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Assesment

$H_0 = \mu_1 = \mu_2$
$H_1 = \mu_1 \neq \mu_2$

Q: 1 Data

$\bar{x}_1 = 89$, $S_1 = 4$, $n_1 = 50$

$\bar{x}_2 = 82$. $S_2 = 9$, $n_2 = 120$

Formula

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$= \frac{89 - 82}{\sqrt{\frac{4^2}{50} + \frac{9^2}{120}}}$$

$$= \frac{7}{\sqrt{\frac{16}{50} + \frac{81}{120}}}$$

$$= \frac{7}{\sqrt{0.32 + 0.675}} = \frac{7}{\sqrt{0.995}}$$

$$= 7.02$$

Los = 0.05 = t - Critical = 1.96.

$7.02 > 1.96$

H_0 Reject

Q:2 Data (O).

Category	Canser	No Canser	Total
Smokers	220	230	550
Non-smokes	350	640	990
Total.	680	910	1590

(E) = Expected

$$E = \frac{(\text{Row Total} \times \text{Column Total})}{\text{Grand total}}$$

Smokes & Cancer

$$E = \frac{550 \times 680}{1590} = 235.22$$

Smokers & Non cancer

$$E = \frac{550 \times 910}{1590} = 314.78$$

Non-Smokers & Cancer:

$$E = \frac{990 \times 680}{1590} = 444.78$$

Non Smokers & No Cancer:

$$E = \frac{990 \times 910}{1590} = 565.22$$

→ Chi-square formula

$$x^2 = \frac{\Sigma (O-E)^2}{E}$$

$$x^2 = \frac{(220-235.22)^2}{235.22} + \frac{(230-314.78)^2}{314.78} + \frac{(350-444.78)^2}{444.78}$$

$$+ \frac{(640-565.22)^2}{565.22}$$

$$x^2 = 0.98 + 22.84 + 20.20 + 9.90$$

$$53.92$$

→ Decision

x^2 Calculated = 53.92

x^2 Critical = 3.84

$53.92 > 3.84$

→ Smoking and cancer are not independent
Smoking is associated with cancer.